Curriculum vitae



Dr. ABHIJIT DEY
Assistant Professor
Mechanical Engineering Department
National Institute of Technology Jamshedpur
(An Institution of National Importance)
Ministry of Education, Government of India
Jharkhand, India-831014
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Biosketch

Dr. Abhijit Dey is a dedicated researcher and educator in mechanical engineering, currently working as an Assistant Professor in the Department of Mechanical Engineering at NIT Jamshedpur. He completed his B.Tech from NIT Agartala in 2011, followed by an M.Tech and Ph.D. in 2014 and 2017, respectively. Over the past eight years, he has taught at esteemed institutions such as NIT Silchar, MITS Madanapalle, and NIT Srinagar before joining NIT Jamshedpur in 2024. Dr. Dey is deeply involved in research, with a strong focus on developing thin films, coatings, and ceramic matrix composites (CMCs) for applications in biomedical and aerospace engineering. His expertise also extends to sensor materials and radar-absorbing materials development for specialized industrial uses. Additionally, he has contributed significantly to the study of the machinability of advanced ceramics, CMCs, and MMCs. With an extensive publication record, Dr. Dey has authored more than 80 research papers in peer-reviewed journals and conferences. He also holds over nine patents, industrial designs, and trademarks. At present, he supervises Nine Ph.D. scholars, two of whom have successfully completed their research, along with guiding five M.Tech theses. His leadership extends to five ongoing research projects as a Principal Investigator (PI) or Co-PI, and he has successfully completed four consultancy projects focused on net-zero energy solutions using geothermal approaches. Beyond his research and teaching, Dr. Dey actively contributes to the academic community as a guest editor for three reputed peer-reviewed journals and regularly reviews articles for several international journals. His dedication to advancing materials science and engineering continues to impact both academia and industry.

Research publications (Last 5 Year)

A: International Journals

1. Das, Anshuman; Patel, Saroj Kumar; Arakha, Manoranjan; Dey, Abhijit; Biswal, Bibhuti Bhushan; "Processing of hardened steel by MQL technique using nano cutting fluids"; Materials and Manufacturing Processes; 2021; 36(3): 316-328; Taylor & Francis.

2. Khan, Mohammad Mohsin; Dey, Abhijit; "Selection of optimal processing condition during abrasive wear of in-situ ZA-37/TiCp composites using MCDM technique"; Ceramics International; 2021; 47(14): 20090-20107; Elsevier.

3. Dey, Abhijit; Pandey, Krishna Murari; Rai, Ram Naresh; Khan, Akhtar; Das, Anshuman; "Application potential of response surface method on electro discharge machining of AA6061-Cenosphere AMCs prepared by compocasting method"; Surface Review and Letters; 2021; 28(7): 2150056; World Scientific Publishing Company.

4. Khan, Mohammad Mohsin; Dey, Abhijit; Hajam, Mohammad Irfan; "Experimental investigation and optimization of dry sliding wear test parameters of aluminum based composites"; Silicon; 2022; 14(8): 4009-4026; Springer Netherlands Dordrecht.

5. Dey, Abhijit; Khan, Mohammad Mohsin; "MCD approach for assessing the optimal operating condition during sliding wear characteristics of LM25/SiC AMCs processed by an improved stir casting technique"; Silicon; 2022; 14(7): 3673-3691; Springer Netherlands.

6. Dey, Abhijit; Shrivastav, Mrinal; Kumar, Pawan; "Optimum performance evaluation during machining of Al6061/cenosphere AMCs using TOPSIS and VIKOR based multi-criteria approach"; Proceedings of the Institution of Mechanical Engineers Part B: Journal of Engineering Manufacture; 2021; 235(13): 2174-2188; SAGE Publications Sage UK: London, England.

7. Khan, Mohammad Mohsin; Dey, Abhijit; "Hybrid MCDM approach for examining the high-stress abrasive wear behaviour of in situ ZA-27/TiCp MMCs"; Materials Chemistry and Physics; 2022; 277: 125319; Elsevier.

8. Lal, Basant; Dey, Abhijit; Wani, Mohamamd Farooq; "Prediction of surface roughness during dry sliding wear: Characteristics of Ti-6Al-4V alloys"; International Journal of Surface Engineering and Interdisciplinary Materials Science (IJSEIMS); 2022; 10(1): 01-Dec; IGI Global Scientific Publishing.

9. Dey, Amit Kumar; Dey, Abhijit; "Selection of optimal processing condition during removal of methylene blue dye using treated betel nut fibre implementing desirability-based RSM approach"; Response Surface Methodology in Engineering Science; 2021; IntechOpen.

10. Goswami, Rumi; Dey, Amit Kumar; Dey, Abhijit; "Positive impact on environment due to COVID-19 lockdowns in parts of India: A review"; Environmental Engineering & Management Journal (EEMJ); 2022; 21(4).

11. Dey, Amit Kumar; Dey, Abhijit; Goswami, Rumi; "Fixed-bed column analysis for adsorption of Acid Scarlet 3R dye from aqueous solution onto chemically modified betel nut husk fibre"; Desalination and Water Treatment; 2022; 252: 381-390; Elsevier.

12. Dey, Amit Kumar; Dey, Abhijit; Goswami, Rumi; "Adsorption characteristics of methyl red dye by Na2CO3-treated jute fibre using multi-criteria decision making approach"; Applied Water Science; 2022; 12(8): 179; Springer International Publishing Cham.

13. Khan, Mohammad Mohsin; Dey, Abhijit; "Microstructure-abrasive wear correlation of in situ ZA27/TiC composites"; Ceramics International; 2022; 48(16): 22928-22942; Elsevier.

14. Lal, Basant; Dey, Abhijit; Wani, Mohammad Farooq; "Influence of temperature and applied load on adhesion and abrasion wear characteristics of Ti6Al4V alloy"; International Journal of Surface Engineering and Interdisciplinary Materials Science (IJSEIMS); 2022; 10(1): 01-Nov; IGI Global.

15. Dey, Abhijit; Rathee, Premdeep Singh; Khan, Mohammad Mohsin; "Computational fluid dynamics approach for frictional heat flow analysis during friction stir welding process"; Arabian Journal for Science and Engineering; 2023; 48(3): 3749-3763; Springer Berlin Heidelberg.

16. Lal, Basant; Dey, Abhijit; Wani, MF; "Effect of frictional heating on mechanically mixed layers during dry sliding contact of Ti-6Al-4V alloys at high temperature in vacuum condition"; Surface Review and Letters; 2022; 29(10): 2250127; World Scientific Publishing Company.

17. Goswami, R; Dey, AK; "Cationic dye removal using surface treated activated carbon as an adsorbent"; Environmental Science: Water Research & Technology; 2022; 8(11): 2545-2566; Royal Society of Chemistry.

18. Dey, Amit Kumar; Dey, Abhijit; Goswami, Rumi; "Selection of optimal performance characteristics during adsorption of methyl red dye using sodium carbonate treated jute fibre"; Desalination and Water Treatment; 2022; 260: 187-202; Elsevier.

19. Mahilary, HK; Dey, AK; "Preparation and application of carboxylated and mechanically attrited carbon for adsorptive removal of crystal violet dye"; Environmental Science: Water Research & Technology; 2023; 9(3): 861-882; Royal Society of Chemistry.

20. Chaudhary, G; Dey, A; Dey, AK; "Stress and delamination analysis on overlapped-grouped and overlapped-dispersed tapered composite laminate"; International Applied Mechanics; 2022; 58(5): 613-622; Springer US New York

21. R Bharti, MM Butt, A Dey, "Dynamic Emissivity Control in MXene-Based Materials for Next-Generation Thermal Camouflage," Journal of Porous Materials, 2024, pp. 1-25.

22. B Lal, A Dey, MF Wani, "Influence of TiCp Ceramic Reinforcement on Ti6Al4V Alloy Matrix via Spark Plasma Sintering Technique," Advances in Materials and Processing Technologies, 2024, 11(1), pp. 589-607.

23. B Lal, A Dey, "Advancements in Titanium Nano-Composites: Microstructure and Fretting Wear Resistance via SPS," International Journal of Minerals, Metallurgy and Materials, 2024.

24. Z Mukhtar, A Dey, N Kundan, "Optimized Surface Engineering of Ti-6Al-4V: Comprehensive Coating Evaluation for Biomedical Applications," Surfaces and Interfaces, 2024, 56, 105735.

25. B Lal, A Dey, U Maurya, A Ali, "Morphology of Wear Debris During Dry Sliding of Ti-6Al-4V Alloy at High Temperature Under Vacuum Environments," Surface Review and Letters, 2024, SEPT, 2550044.

26. B Lal, A Dey, U Maurya, A Ali, "Erratum—Morphology of Wear Debris During Dry Sliding of Ti–6Al–4V Alloy at High Temperature Under Vacuum Environments," Surface Review and Letters, 2024, 2492002.

27. R Bharti, M Mursaleen, A Dey, "Optimizing Stealth Thin Films: Nanoindentation and Nanoscratch Testing for Enhanced Mechanical Properties," Proceedings of the Institution of Mechanical Engineers, Part E: Journal of …, 2024.

28. KA Sheikh, MM Khan, A Dey, "Optimizing Wear Resistance of ZA-27 Alloy Through Analytical Modelling: A Study on the Effect of Titanium Carbide Content, Load, and Speed," Proceedings of the Institution of Mechanical Engineers, Part J: Journal of …, 2024.

29. B Lal, A Dey, U Maurya, A Ali, "Morphology of Wear Debris During Dry Sliding of Ti6Al4V/TiC Composite at Elevated Temperatures in Vacuum Conditions," Surface Review and Letters, 2024, 2550044.

30. B Lal, A Dey, "Effect of TiC Nano-Particle on Microstructure Evolution and Tribological Behaviour of Ti6Al4V Composites Fabricated via Spark Plasma Sintering," Tribology International, 2024, 197, 109775.

31. Z Mukhtar, N Kundan, A Dey, "Corrosion and Wear Characterization of Ti6-Al-4V Alloy: Experimental Analysis and Performance Evaluation," Tribology International, 2024, 197, 109745.

32. B Lal, A Dey, MF Wani, "Nano-Indentation Characterization of TiC Nano Ceramic Reinforced Ti-6Al-4V Matrix Composite," Journal of Materials Engineering and Performance, 2024, pp. 1-14.

33. A Dey, B Lal, AK Dey, SK Jain, "Development of Optimum Performance Model During Spark-Eroded Processing of Rheocasted Al6061/Cenosphere MMCs," Advances in Materials and Processing Technologies, 2024, pp. 1-14.

34. R Bharti, MM Butt, A Dey, "Advancements in Electromagnetic Shielding Coatings for Aerospace Applications: A Comprehensive Review," Surface Innovations, 2024, 12(8), pp. 440-450.

35. KA Sheikh, MM Khan, A Dey, "Comprehensive Analysis of TiC-Reinforced ZA-37 Alloy Matrix Composites: Microstructural, Hardness, and Corrosion Performance," Journal of Bio-and Tribo-Corrosion, 2024, 10(2), 19.

36. B Lal, A Dey, "Fretting Wear Characteristics of Ti6Al4V/TiC Nano Composites Using Response Surface Method," Tribology International, 2024, 193, 109412.

37. A Dwivedi, MM Khan, A Dey, MI Hajam, "Computational Analysis of the Effect of Flow Parameters on Friction Factor and Thermal Performance of an Artificially Broken Arc Rib Solar Air Heater," International Journal of Computational Materials Science and Engineering, 2024, 13.

38. Z Mukhtar, A Dey, N Kundan, FA Mir, "Advancements in Biomedical Coatings: A Comprehensive Review of DC Magnetron Sputtering on Ti–6Al–4V Alloy," Proceedings of the Institution of Mechanical Engineers, Part J: Journal of …, 2024.

39. AKD A Dey, "Use of Electrochemical Technique for the Removal of Lead and Nickel from Modeled Aqueous Solution," European Chemical Bulletin, 2024.

40. A Dey, PS Rathee, MM Khan, "Computational Fluid Dynamics Approach for Frictional Heat Flow Analysis During Friction Stir Welding Process," Arabian Journal for Science and Engineering, 2024, 48(3), pp. 3749-3763.

41. HK Mahilary, AK Dey, "Preparation and Application of Carboxylated and Mechanically Attrited Carbon for Adsorptive Removal of Crystal Violet Dye," Environmental Science: Water Research & Technology, 2024, 9(3), pp. 861-882.

42. A Dey, MM Khan, "Investigation of Erosion Wear Characteristics of LM13/SiC Composites Implementing Box-Behnken Design Methodology," Journal of Bio-and Tribo-Corrosion, 2024, 8(4), 90.

43. MM Khan, A Dey, Z Rubanee, K Mushtaq, MI Hajam, SSU Islam, MB Shah, "Evaluation of Microstructure, Mechanical Properties, and Biocompatibility of Biodegradable Zinc-Based Alloys for Implants," International Conference on Advances in Materials and Manufacturing, 2024, pp. 267-277.

44. B Lal, A Dey, MF Wani, "Effect of Frictional Heating on Mechanically Mixed Layers During Dry Sliding Contact of Ti-6Al-4V Alloys at High Temperature in Vacuum Condition," Surface Review and Letters, 2024, 29(10), 2250127.

45. A Das, SR Das, JP Panda, A Dey, KK Gajrani, N Somani, NK Gupta, "Machine Learning-Based Modeling and Optimization in Hard Turning of AISI D6 Steel with Advanced AlTiSiN-Coated Carbide Inserts to Predict Surface Roughness and Other Machining …," Surface Review and Letters, 2024, 29(10), 2250137.

46. R Bharti, M Mursaleen, A Dey, “Carbon-based radar-absorbing materials: A comprehensive review of design, properties, and applications,” *Radiation Effects and Defects in Solids*, **2025**, 1–24.

47. R Bharti, M Mursaleen, A Dey, "Microstructure evolution and optical–infrared properties of Ag/TiN/Al_2O_3 multilayers deposited by magnetron sputtering," *Canadian Metallurgical Quarterly*, **2025**, 1–11.

48. R Bharti, M Mursaleen, A Dey, "Preparation of Ag+TiN/TiN/Al_2O_3 multilayer coatings by magnetron sputtering and their optical properties," *Technical Physics*, **2025**, 1–6.

49. R Bharti, M Mursaleen, A Dey, "Thermal and optical performance of Ag+TiN–TiN–Al_2O_3 multilayer coatings for infrared camouflage in aerospace applications," *Micro and Nanostructures*, **2025**, 208359.

50. A Dey Rohit Bharti, Mohammad Murseen Bhatt, "Advanced multilayer thin films for stealth applications: Optical, electrical, and thermal enhancements—A comprehensive study," *Journal of Optics*, **2025**, https://doi.org/10.1007/s12596-025-02919.

51. A Dey Gourav Chaudhary, Amit Kumar Dey, "Delamination prediction on staircased-dispersed tapered laminate composite," *International Applied Mechanics*, **2025**, 61(4).

52. A Dey, SM Hazarika, "Object stiffness discrimination through dynamic time warping and multi-criteria decision-making," *International Journal of Intelligent Robotics and Applications***, 2025**, 1–17.

53. U Iqbal, NZ Khan, A Dey, "Friction and wear characterization of ZE41 magnesium alloy for medical implants," *Proceedings of the Institution of Mechanical Engineers, Part J: Journal of Engineering Tribology*, **2025**.

54. B Lal, A Dey, MF Wani, "Nano-indentation characterization of TiC nano ceramic reinforced Ti-6Al-4V matrix composite," *Journal of Materials Engineering and Performance*, **2025**, 34(11), 10577–10590.

55. A Dey, MM Khan, "Optimizing WEDM parameters for aluminum metal matrix composites using hybrid PSO–TOPSIS approach," *Journal of Advanced Manufacturing Systems*, **2025**.

56. ZA Ahanger, SS Saleem, A Dey, "Physical vapour deposition technologies of functionally graded coatings," *Proceedings of the Institution of Mechanical Engineers, Part L: Journal of Materials: Design and Applications*, **2025**.

57. B Lal, MF Wani, A Dey, A Singh, "In-vitro analysis of TC4 alloy reinforced with inert ceramics for biomedical applications synthesized using spark plasma sintering," *Tribology International*, **2025**, 110744.

58. Z Mukhtar, N Kundan, A Dey, "Development of functionally graded films and coatings for biomaterials: Methods, properties, and challenges—A comprehensive review," *Tribology International*, **2025**, 110714.

59. KA Sheikh, MM Khan, A Dey, MF Wani, “Advancements in surface engineering: A comprehensive review of functionally graded coatings for biomaterial implants,” *Results in Surfaces and Interfaces*, **2025**, 100509.

60. A Dey, B Lal, AK Dey, SK Jain, “Development of optimum performance model during spark-eroded processing of rheocasted Al6061/cenosphere MMCs,” *Advances in Materials and Processing Technologies*, **2025**, 11(2), 1215–1228.

61. R Bharti, MM Butt, A Dey, “Dynamic emissivity control in MXene-based materials for next-generation thermal camouflage,” *Journal of Porous Materials*, **2025**, 32(2), 399–423.

62. R Bharti, M Mursaleen, A Dey, “Optimizing stealth thin films: Nanoindentation and nanoscratch testing for enhanced mechanical properties,” *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*, **2025**.

63. B Lal, A Dey, MF Wani, “Influence of TiCp ceramic reinforcement on Ti-6Al-4V alloy matrix via spark plasma sintering technique,” *Advances in Materials and Processing Technologies*, **2025**, 11(1), 589–607.

64. B Lal, A Dey, “Advancements in titanium nano-composites: Microstructure and fretting wear resistance via SPS,” *International Journal of Minerals, Metallurgy and Materials*, **2025**.

65. Z Mukhtar, A Dey, N Kundan, “Optimized surface engineering of Ti-6Al-4V: Comprehensive coating evaluation for biomedical applications,” *Surfaces and Interfaces*, **2025**, 56, 105735.

66. R Bharti, MM Butt, A Dey, “Engineering broadband electromagnetic wave absorbers: Innovations in carbon-based composites, magnetic alloys, and impedance matching,” Russian Journal of Non-Ferrous Metals, **2025**, 66(6), 414–432.

67. MQU Ain, A Dey, MF Wani, R Sehgal, MM Khan, “Microstructural and mechanical properties of nickel alloy C-263 for nuclear applications,” Canadian Metallurgical Quarterly, **2025**, 1–12.

68. R Bharti, M Mursaleen, A Dey, “Fabrication and optical–infrared performance of Ag+TiN/TiN/Al_2O_3 multilayer coatings by multi-target magnetron sputtering,” Canadian Metallurgical Quarterly, **2025**, 1–15.

69. R Bharti, M Mursaleen, A Dey, “Microstructural and optical–infrared properties of Ag–TiN–Al_2O_3 multilayers fabricated by magnetron sputtering,” Ceramics International, **2026**, 52(8), 10244–10254.

70. Z Mukhtar, A Dey, N Kundan, FA Mir, “Advancements in biomedical coatings: A comprehensive review of DC magnetron sputtering on Ti–6Al–4V alloy,” Proceedings of the Institution of Mechanical Engineers, Part J: Journal of Engineering Tribology, **2026**, 240(3). doi:10.1177/13506501241282751.

71. Z Mukhtar, A Dey, N Kundan, “Influence of substrate temperature on the structural and mechanical behavior of functionally graded TaN/CrN/ZrN coatings on Ti–6Al–4V,” Results in Surfaces and Interfaces, **2026**, 23, 100784.

72. R Bharti, M Mursaleen, A Dey, “Microstructure and thermal-infrared performance of Ag/TiN/Al_2O_3 multilayer thin films,” Photonics and Nanostructures – Fundamentals and Applications, **2026**, 71, 101533.

73. A Dey, Z Mukhtar, N Kundan, “Synthesis and characterization of functionally graded TaN/CrN/ZrN thin films on Ti–6Al–4V,” Surface and Coatings Technology, **2026**, 526, 133338.

74. MIR Q UL Ain, A Dey, MF Wani, R Sehgal, MM Khan, “Functionally graded WC–Cr_3C_2–Ni coatings on Ni-C263: Microstructural design and mechanical reliability,” Physica Scripta, **2026** (online published 12 May 2026).

B: National Journals

1. **Abhijit Dey**, Krishna Murari Pandey, Pannalal Choudhury, **“**Post Buckling Analysis of Circular Ring Stiffened Filament Wound Composite Pressure Vessels by Finite Element Analysis”, Journal of Emerging Technology In Mechanical Science and Engineering, Volume-V, special Issue, March 2014,pp:82-86.
2. Pannalal Choudhury, Gaurav Chaudhary*, **Abhijit Dey**, Krishna Murari Pandey, **“**Interlaminar Stress Analysis of Composite Laminates with External Ply-Drop using Finite Element Method”, Journal of Emerging Technology In Mechanical Science and Engineering, Volume-V, special Issue, March 2014, pp:75-81.

C: International Conferences

1. **Abhijit Dey**, Gaurav Chaudhary, Pannalal Choudhury, Krishna Murari Pandey, “Study the Effect of Buckling Pressure on End Shortening of Circular Ring Stiffened Composite Pressure Vessels through Finite Element Analysis”, International Conference on "Contemporary Challenges in Management, Technology & Social Sciences”, at MGIMT, Lucknow on April 05-06, 2014,ISBN:978-81-928926-3-4,pp:403-409.
2. **Abhijit Dey** and Krishna Murari Pandey, ‘Machinability study of compocasted cenosphere/AA6061 aluminium matrix composites using grey based response surface methodology (GRSM)’, ICCE-24 24th Annual International Conference on Composites and Nano Engineering, 17 Jul 2016 - 23 Jul 2016, Sanya, Hainan Island, China.
3. Gaurav Chaudhary, **Abhijit Dey**, Amit kumar Dey and Subhankar Das, “Delamination Analysis of ply Drop-off in Tapered Laminated Composite”, International Conference on Modern Technology in Engineering and Science (ICMTES) 2017 at KIRET Andhra Pradesh, India, 2nd -3rd june 2017.
4. Amit kumar Dey, **Abhijit Dey**, and Mawfungra Brahma, “Design of Overhead Tank for the Newly Built Guest House in CIT Assam”, International Conference on Modern

Technology in Engineering and Science (ICMTES) 2017 at KIRET Andhra Pradesh, India, 2nd -3rd june 2017.

5. **Abhijit Dey**, Krishna Murari Pandey, Biplab Das, "optimisation of wire electro discharge machining parameter using desirability based multi objective rsm while machining of al6061–cenosphere amcs", Proceedings of SEEP 2017, 27-30 june 2017, Bled, Slovenia. https://doi.org/10.18690/978-961-286-057-8.5.

D: National Conferences

Abhijit Dey, Pannalal Choudhury, Krishna Murari Pandey, "A comparison study of filament wound composite cylindrical shell used in under water vehicle application by Finite Element Method", accepted in 5th International & 26th National Conference on All India Manufacturing Technology, Design and Research Conference being held at Indian Institute of Technology Guwahati, Assam, India during December 12-14, 2014.

Patents published and granted

Sl.No.	Name Of The Patents	Date & Number	Awarding Organization, published /granted
1.	Biodiesel operating engine	14-11-2022, 374118-001	Indian Patent Office, Granted
2.	Sand jet machining	13-10-2021, 351279-001	Indian Patent Office, Granted
3.	Wi-fi based oxygen concentrator	26 Jul 2021, 346741-001	Indian Patent Office, Granted
4.	Ein system zur herstellung eines nanobrennstoffs	23 Jan 2023, 20 2022 100127.8	German Patent Office, Granted
5.	System zur hohen adsorptiven entfernung vonkationischem farbstoff unter verwendung von mitanionischen tensiden modifizierter aktivkohle	30 Mar 2023,20 2023 100304.4	German Patent Office, Granted
6.	A composition and a method of preparing an adsorbent for removing a cationic dye from an aqueous solution	2023, 02840	South African Patent Office, Granted
7.	A connector assembly device and a system to prolong the life of leds	14/3/2025, 202531019695A	INDIAN PATENT OFFICE, Published
8.	Hand-Operated Chain Link Bending Fixture with Rotary Bearing Assembly	476908-001	Granted
9.	Manual Chain Link Bending Tool with Dual Levered Roller Mechanism	468460-001	Granted
10.	A Threshing Machine With A Dust Trapping System	202531090634	Published
11.	A Biomedical Implant	202531065599	Published

	Material Comprising TiC Nanoparticle Reinforced TC4 Titanium Alloy and Its Method of Preparation		

Ongoing Project

Title of the Project	**Sponsoring Agency**	**Project Duration [in months]**	**PI/ Co Investigator**	**Completed/ Ongoing**
Development of Novel WC-Cr2C3-Ni functionally graded composite coatings on Nickel based C-263 super alloy for Nuclear application	SERB-CRG	36	PI	Ongoing
Design and Development of Novel TiN-CrN-ZrN functionally graded composite coatings for biomedical implant material	SERB-CRG	36	Co-PI	Ongoing
"Effect of Laser surface texturing on the biodegradability of zinc based implant material"	(/JKST&IC/SRE/275-80)	24	PI	Ongoing

Details of conference organized

Name of the Conference	**Responsibility**	**Start Date**	**Start Date**	**Sponsoring Agency**
2nd International Indo	Coordinator	17 june 19	21 june 19	DST & TEQIP

TunisianSymposium on' 'Tribology for Sustainability				III
Triboindia-2023	Organising committee member	05 Oct 2023	07 Oct 2023	Tribology Society of India

H. Research Interest

- Advanced materials Processing and Characterizations
- Non conventional machining
- Thinfilm & Coatings

PhD Guidance

1. Mr. Bashant Lal (16-3-02-103), High Temperature Dry sliding Wear Behaviour of Titanium alloys under Vaccum environment for Biomedical Application.

2. Mr. Khurshad Ahmad (2021PHAMEC007), 'Investigation on Fabrication and characterization of Insitu-TiC reinforced Zinc Aluminum alloys'.

PG Guidance

1. Abhay Kumar (2018MMECIT007), Condition Monitoring System for Solar Power Plants with Sensors and Decision-Making Method for Panel Selection.

2. Mr. Premdeep Singh (2019MMECMS010) A Microscopic Approach On Frictional Heat Flow Analysis of 6061-T6 Aluminum Alloy During Friction Stir Welding Process Using CFD.

3. Mr. **Gibran Hussain (2020MMECMS004)** Machinability Study of Titanium carbide Reinforced ZA37alloy prepared by Advanced stir-casting process.

Courses Taught

A: Under Graduate

- Mechanics of Solid
- Manufacturing Processes
- Machine Drawing
- Engineering Graphics & Computer modeling
- Industrial Engineering-l

B: Post Graduate

- Advanced Manufacturing Processes

Workshop/STTP Organized

- One Week Short-Term Training Program on **"Additive Manufacturing from 3D Printing to the Factory Floor"** from **August 31- 4th September, 2020**, Department of Mechanical Engineering, National Institute of Technology Srinagar under TEQIP - III, Coordinator: Dr. **Abhijit Dey & Dr. M.M. Khan.**

- Five-days short term course on "**Computational Fluid Dynamics**" Scheduled from 17-**22 Aug 2020** in the Department of Mechanical Engineering at NIT Srinagar under TEQIP - III, Coordinator: Dr. **Abhijit Dey & Dr. H.S. Pali.**

- One Week Short-Term Training Program on **"ADVANCED MATERIALS PROCESSING AND CHARACTERIZATION"** from **22nd - 26th July, 2019**, Department of Mechanical Engineering, National Institute of Technology Srinagar under TEQIP - III, Coordinator: Dr. **Abhijit Dey & Dr. M.D.Sharma.**

- **2ND INTERNATIONAL INDO-TUNISIAN SYMPOSIUM ON 'TRIBOLOGY FOR SUSTAINABILITY'** from **17th - 21st June, 2019**, Department of Mechanical Engineering, National Institute of Technology Srinagar under TEQIP - III, Coordinator: Dr. **Abhijit Dey, Dr. M.D. Sharma & Dr. M. F. Wani.**

- **Three days Short Term Course on 'Nano-Physics', from 06th April 2019 – 08th April 2019,** Department of Mechanical Engineering, National Institute of Technology Srinagar under Twinning Activity, Coordinator: Dr. **Abhijit Dey.**

- **Three days Short Term Course on "Computer Programming", from 17th May 2019 – 19th May 2019,** Department of Mechanical Engineering, National Institute of Technology Srinagar under Twinning Activity, Coordinator: Dr. **Abhijit Dey.**

- One Day Short term Program on 'THE FASCINATING WORLD OF MATERIALS' held on 27th May 2019 in the Department of Mechanical Engineering, National Institute of Technology Srinagar under TEQIP - III, Coordinator: Dr. **Abhijit Dey, Dr. M.M. Khan.**

Workshop/seminar attended:

1.	1Week Workshop on, "**Computational Mechanics and Modelling (CMM 2016)".**	NIT Silchar Under TEQIP-II	November 12-16, 2016
2.	1 week Workshop on, "**Computational Combustion"**	NIT Silchar Under TEQIP-II	31st October to 5 th November 2016
3.	1 week Workshop on, **"Advanced Joining Technologies"**	NIT Silchar Under TEQIP-II	Oct 17-21, 2016
4.	GIAN course on "**Surface Engineering**"	NIT Silchar Under **MHRD**	Sept13-17, 2016

5.	1 Week short term course on**, "application of probability theories optimization techniques".**	NIT Silchar Under TEQIP-II	Dec 09-13, 2013
6.	3 Days short term course on, "**Application of Soft Computing Technique in Engineering**".	NIT Silchar Under TEQIP - II	Nov,9-11,2013
7.	3 Days workshop on, "Advanced **Numerical Techniques Using Mat lab**".	NIT Agartala Under TEQIP - II	Oct 07-09 ,2013
8.	1 Week short term course on, "Failure **Analysis of Engineering Materials"**.	NIT Silchar Under TEQIP - II	Sept 23-27, 2013
9.	1 Week short term course on, "**Joining of Materials".**	NIT Silchar Under TEQIP - II	August 9-13, 2013
10.	1 Day Awareness programme on, "**Industry Oriented Piping Engineering".**	NIT Silchar Under TEQIP - II	2nd Feb, 2013
11.	3 Days workshop on, "Personality **development, Career Building, Website Designing & Mini project Executions**".	NIT Agartala, Dept. Of Mechanical Engg	May 19-21, 2009

Membership

- Life Member - The Society of Polymer Science, India.
- Member - The Institute of Engineers (India).
- Member - ASME.

Reviewer

- ***Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture.***
- ***Journal of The Institution of Engineers (India): Series C***
- ***Clean Technologies and Environmental Policy.***
- ***Tribology International***
- ***Materials and Manufacturing Processes***
- ***Ceramic International***
- ***Sadhana***
- ***Advances in Space Research***
- ***Surfaces and Interfaces***
- ***Results in Surfaces and Interfaces***

- *Carbon*
- *Surface and Coating Technology*
- *Next Materials*

Administrative Responsibilities

1. Twinning In-Charge, Institute from 03.04.2019 – 12.12.2024
2. Member – NIRF, from 01.09.2020- 12.11.2024.
3. Member- NBA, Department, from 03.07.2019- 12.11.2024.
4. **NIRF convener**, Institute from dec. 2025 to till date

Equipment's utilized

EDM, WDEM, SEM, FESEM, EDX, XRD, Micro Hardness Tester, UTM, High resolution optical Microscope (Taylor Hobson-80G), 3D optical Surface roughness Tester, Stir casting Setup with control environments, Induction Furnace, Muffle Furnace.

Optimization Process used

Gray relation analysis(GRA), Response Surface Methodology (RSM), GRSM.

Software Skills: SOLID WORKS**,** AUTO CAD 4-10, CATIA V5, ANSYS APDL 12.0, 14.0.

Personal Details

Date of birth **:** 20.09.1988
Gender **:** Male.
Father's Name **:** Sri Subhash Chandra Dey.
Mother's Name **:** Smt. Bina Paul (Dey).
Permanent Address **:** Vill. Sovapur, Po. Rabindra nagar, Sonamura, Sepahijala, Tripura, 799131.
Marital Status **:** Married
Nationality **:** Indian
Areas of Interest **: Academics**: Strength of Material, advance manufacturing processes, production technology.

Hobbies : **Sports-** Cricket, Carom.
Others- Listening music, reading story books, net surfing, Natural Photography.

REFERENCES

1. **Dr. K M Pandey**, Ph.D.(IITK)
   Professor and Ex-Head,

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Please visit: http://www.iitg.ernet.in/aimtdr2014/

DECLARATION:

I hereby declare that the above written particulars are true to the best of my knowledge and belief.



Yours faithfully,

Abhijit Dey.